1 Limits

Let's compare the behavior of the functions

$$f(x,y) = \frac{\sin(x^2+y^2)}{x^2+y^2} \quad \text{and} \quad g(x,y) = \frac{x^2-y^2}{x^2+y^2}$$

as $(x,y) \to (0,0)$. Note that neither function is defined at $(0,0)$:

TABLE 1 Values of $f(x,y)$

x \ y	−1.0	−0.5	−0.2	0	0.2	0.5	1.0
−1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455
−0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
−0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0	0.841	0.990	1.000		1.000	0.990	0.841
0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455

TABLE 2 Values of $g(x,y)$

x \ y	−1.0	−0.5	−0.2	0	0.2	0.5	1.0
−1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000
−0.5	−0.600	0.000	0.724	1.000	0.724	0.000	−0.600
−0.2	−0.923	−0.724	0.000	1.000	0.000	−0.724	−0.923
0	−1.000	−1.000	−1.000		−1.000	−1.000	−1.000
0.2	−0.923	−0.724	0.000	1.000	0.000	−0.724	−0.923
0.5	−0.600	0.000	0.724	1.000	0.724	0.000	−0.600
1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000

It looks like $\lim_{(x,y)\to(0,0)} f(x,y) = 1$ (since the values of the function approach 1 from all directions) while $\lim_{(x,y)\to(0,0)} g(x,y)$ does not exist (since the values of the function near $(0,0)$ do not approach any fixed value).

As in single-variable calculus, we say that $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L$ if $f(\mathbf{x})$ can be made "arbitrarily close" to L by taking $\mathbf{x}$ "sufficiently close" to $\mathbf{a}$.

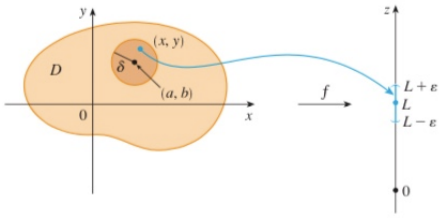


Figure 1: A function $f(\mathbf{x})$ has limit L as $\mathbf{x} \to \mathbf{a}$ if $f(\mathbf{x})$ can be made arbitrarily close to L by taking $\mathbf{x}$ sufficiently close to $\mathbf{a}$.

Here is the precise definition.

Definition 1.1 (Limit)**.** Let U be an open subset of $\mathbb{R}^n$ and $f : U \to \mathbb{R}$, which may or may not be defined at $\mathbf{a} \in U$, and let $L \in \mathbb{R}$. We say that

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L$$

if for every $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that

$$0 < ||\mathbf{x} - \mathbf{a}|| < \delta \implies |f(\mathbf{x}) - L| < \varepsilon.$$

In this case L is called the *limit* of f as $\mathbf{x}$ approaches $\mathbf{a}$.

As in single-variable calculus, another illustration of the definition of a limit can be given in terms of the graph of the function: For any $\varepsilon > 0$, we can find $\delta > 0$ such that if $\mathbf{x} \in B_\delta(\mathbf{a})$ and $\mathbf{x} \neq \mathbf{a}$, then $f(x)$ lies between the horizontal planes $z = L - \varepsilon$ and $z = L + \varepsilon$.

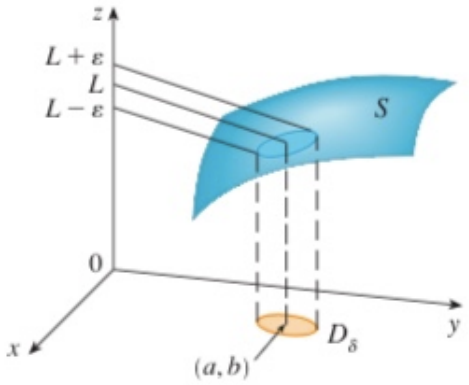


Figure 2: A function f has limit L as $\mathbf{x} \to \mathbf{a}$ if the graph can be bounded between two arbitrarily close planes whenever for all $\mathbf{x} \neq \mathbf{a}$ within an open disk containing centered at $\mathbf{a}$ of sufficiently small radius.

For functions of a single variable, when we let $x \to a$, there are only two directions of approach: from the left or from the right. Recall that $\lim_{x\to a} f(x)$ exists if and only if $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist and are equal (in which case $\lim_{x\to a} f(x)$ is equal to the common value). For functions of two variables, we can let (x,y) approach (a,b) from an infinite number of directions, as long as (x,y) stays within the domain of f.

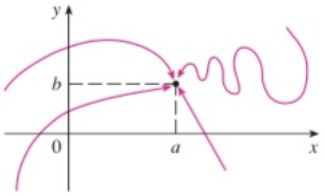


Figure 3: Various paths in the domain of a function $f : \mathbb{R}^2 \to \mathbb{R}$ along which we can approach a point (a,b).

Note that the definition of a limit only depends on the *distance* from (x,y) and (a,b). It does *not* refer to the direction of the approach. Therefore, if the limit exists, then $f(x,y)$ must approach the same limit no matter how (x,y) approaches (a,b). Thus, if we can find two different paths of approach along which the function $f(x,y)$ has different limits, then it follows that $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.

Example 1.2. Consider $\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$. Consider approaching $(0,0)$ along the x-axis. Then

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} = \lim_{x\to 0} \frac{x^2-0}{x^2+0} = \lim_{x\to 0} \frac{x^2}{x^2} = \lim_{x\to 0} 1 = 1.$$

Approaching $(0,0)$ along the y-axis, we find

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} = \lim_{y\to 0} \frac{0-y^2}{0+y^2} = \lim_{y\to 0} \frac{-y^2}{y^2} = \lim_{y\to 0}(-1) = -1.$$

We have found two paths along which $\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ has different values, so $\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist.

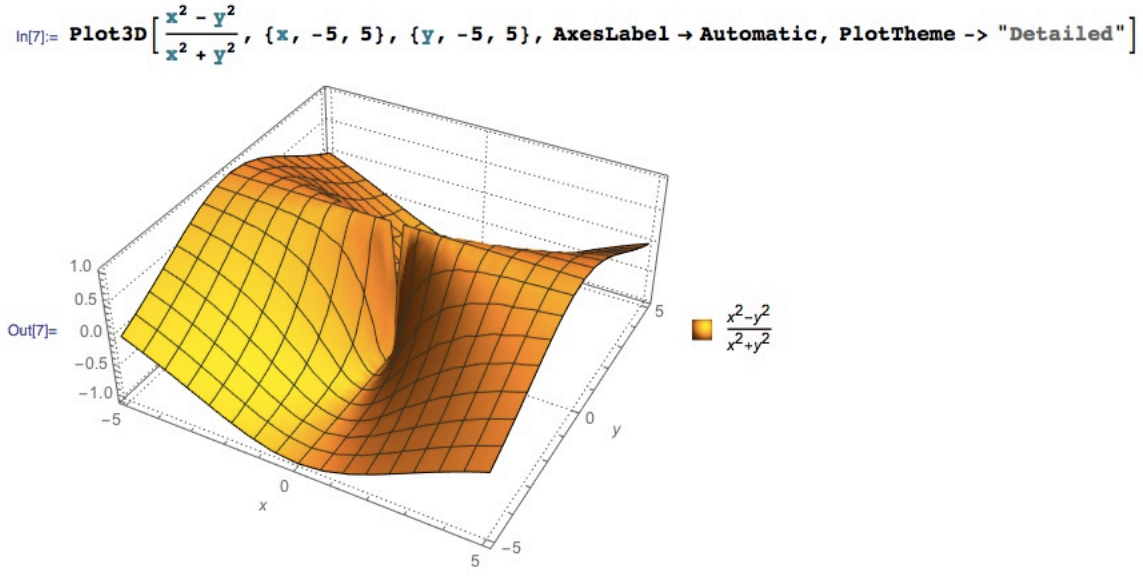


Figure 4: The function $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$ has no limit as $(x,y) \to (0,0)$.

Example 1.3. Consider $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$. Approaching $(0,0)$ along the x-axis gives

$$\lim_{x\to 0} \frac{x\cdot 0}{x^2+0} = \lim_{x\to 0} \frac{0}{x^2} = 0.$$

Approaching $(0,0)$ along the y-axis gives

$$\lim_{y\to 0} \frac{0\cdot y}{0+y^2} = \lim_{y\to 0} \frac{0}{y^2} = 0.$$

While these limits agree, this is *not* sufficient to conclude that $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = 0$, since we must show that we obtain the same limit along *every* path approaching $(0,0)$. Indeed, if we approach $(0,0)$ along the line $y = x$, we obtain

$$\lim_{x\to 0} \frac{x^2}{x^2+x^2} = \lim_{x\to 0} \frac{x^2}{2x^2} = \lim_{x\to 0} \frac{1}{2} = \frac{1}{2}.$$

Thus, $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ does not exist.

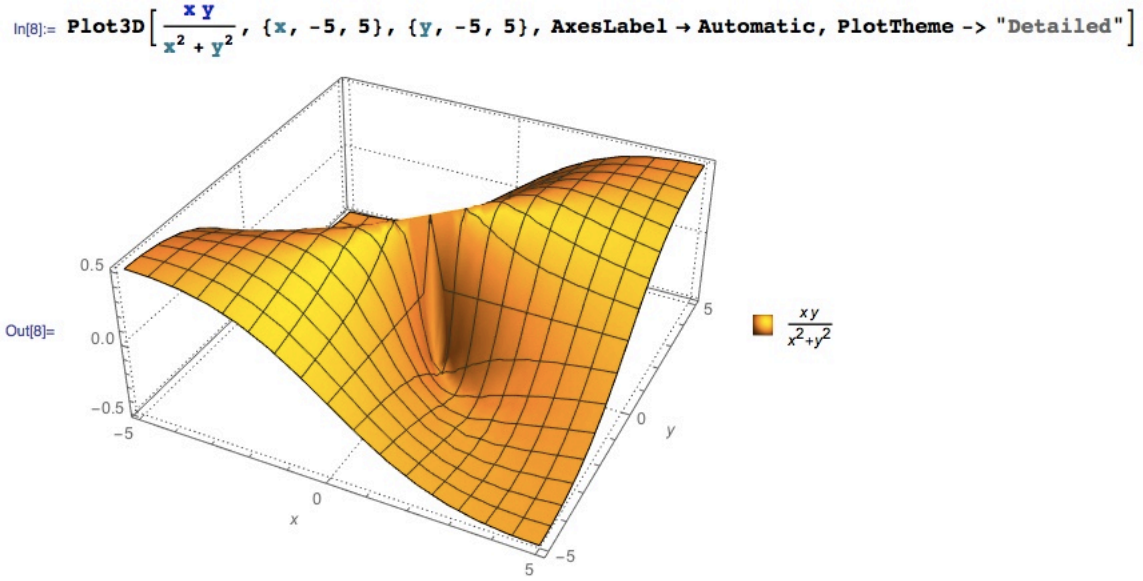


Figure 5: The function $f(x,y) = \frac{xy}{x^2+y^2}$ has no limit as $(x,y) \to (0,0)$.

The previous example shows that it is not sufficient to compute the limit of a multivariable function by restricting the function to paths in the domain, as we would have to check every possible path and there are infinitely many. How, then, does one actually show that a limit exists? There are several techniques, similar to the ones used in single-variable calculus.

Technique 1: Using the definition

The first technique is to use Definition 1.1 to prove that a limit exists.

Here is a useful fact.

Lemma 1.4. *Let* $\mathbf{x} = (x_1, x_2, \ldots, x_n), \mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$. *Then* $|x_i - a_i| \leq ||\mathbf{x} - \mathbf{a}||$ *for all* $i = 1, \ldots, n$.

This lemma is geometrically obvious. For instance, if (x,y) lies in an open disk centered at (a_1, a_2) of radius r, then x must lie in the open interval $(a_1 - r, a_1 + r)$ and y must lie in the open interval $(a_2 - r, a_2 + r)$.

Proof. For each $i = 1, \ldots, n$, $(x_i - a_i)^2 \geq 0$. It follows that, for any fixed i we have

$$\sum_{j \neq i} (x_j - a_j)^2 \geq 0$$

and therefore

$$\sum_{j \neq i} (x_j - a_j)^2 + (x_i - a_i)^2 \geq (x_i - a_i)^2$$

which says $||\mathbf{x}-\mathbf{a}||^2 \geq (x_i-a_i)^2$. Since both sides are non-negative, this implies

$$||\mathbf{x}-\mathbf{a}|| \geq \sqrt{(x_i-a_i)^2} = |x_i-a_i|.$$

□

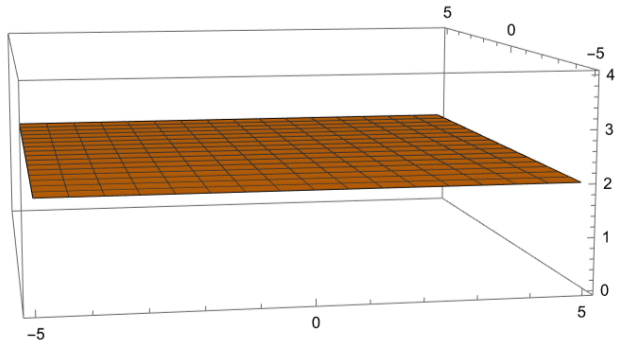
Theorem 1.5. *Let a,b,c be fixed real numbers. Then*

(a) $\lim_{(x,y)\to(a,b)} c = c.$

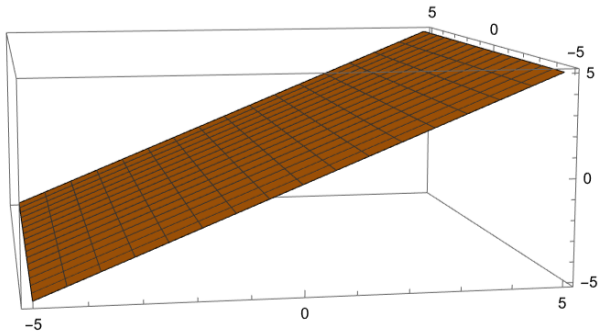
(b) $\lim_{(x,y)\to(a,b)} x = a.$

(c) $\lim_{(x,y)\to(a,b)} y = b.$

Similar limits hold in $\mathbb{R}^n$.



(a) Graph of $z = f(x,y) = c$.



(b) Graph of $z = f(x,y) = x$.

Proof.

(a) Let $\varepsilon > 0$. Since $|c-c| = 0$, by taking δ to be any positive real number, then $0 < \sqrt{(x-a)^2+(y-b)^2} < \delta$ implies $|f(x,y) - f(a,b)| = |c-c| = 0 < \varepsilon$. This proves that $\lim_{(x,y)\to(a,b)} c = c$.

(b) Let $\varepsilon > 0$. We have $|f(x,y) - f(a,b)| = |x-a|$. Since

$$|x-a| = \sqrt{(x-a)^2} \leq \sqrt{(x-a)^2+(y-b)^2},$$

by taking $\delta = \varepsilon$, $0 < \sqrt{(x-a)^2+(y-b)^2} < \delta$ implies that $|f(x,y) - f(a,b)| = |x-a| < \varepsilon$. This proves that $\lim_{(x,y)\to(a,b)} x = a$.

(c) The proof is essentially the same as that of part (b) and is left as an exercise.

□

Exercise 1.6. Prove part (c) of Theorem 1.5.

Proof. Let $\varepsilon > 0$. We have $|f(x,y) - f(a,b)| = |y-b|$. Since

$$|y-b| = \sqrt{(y-b)^2} \leq \sqrt{(x-a)^2+(y-b)^2},$$

by taking $\delta = \varepsilon$, $0 < \sqrt{(x-a)^2+(y-b)^2} < \delta$ implies that $|f(x,y) - f(a,b)| = |y-b| < \varepsilon$. This proves that $\lim_{(x,y)\to(a,b)} y = b$. □

From the proof of the previous exercise, we see that the following is true.

Proposition 1.7. *Suppose f is a function of one variable, say x, defined on an open interval containing a, but perhaps not at a itself, such that $\lim_{x\to a} f(x) = L$. Extend f to a function of n variables by taking $f(x,y,z,\dots) = f(x)$ (that is, f is actually independent of the other $n-1$ variables). Then, if $\mathbf{a} = (a,\dots)$ is any vector whose x-component is a, then*

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L.$$

Proof. Let $\varepsilon > 0$. Since $\lim_{x\to a} f(x) = L$, there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|f(x) - L| < \varepsilon$. It follows that, by taking $0 < ||\mathbf{x}-\mathbf{a}|| < \delta$, we have

$$0 < |x-a| < ||\mathbf{x}-\mathbf{a}|| < \delta$$

and therefore $|f(\mathbf{x}) - L| = |f(x) - L| < \varepsilon$. Thus,

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L.$$

□

Example 1.8. $\lim_{(x,y)\to(\frac{\pi}{2},3)} \sin(x) = 1.$

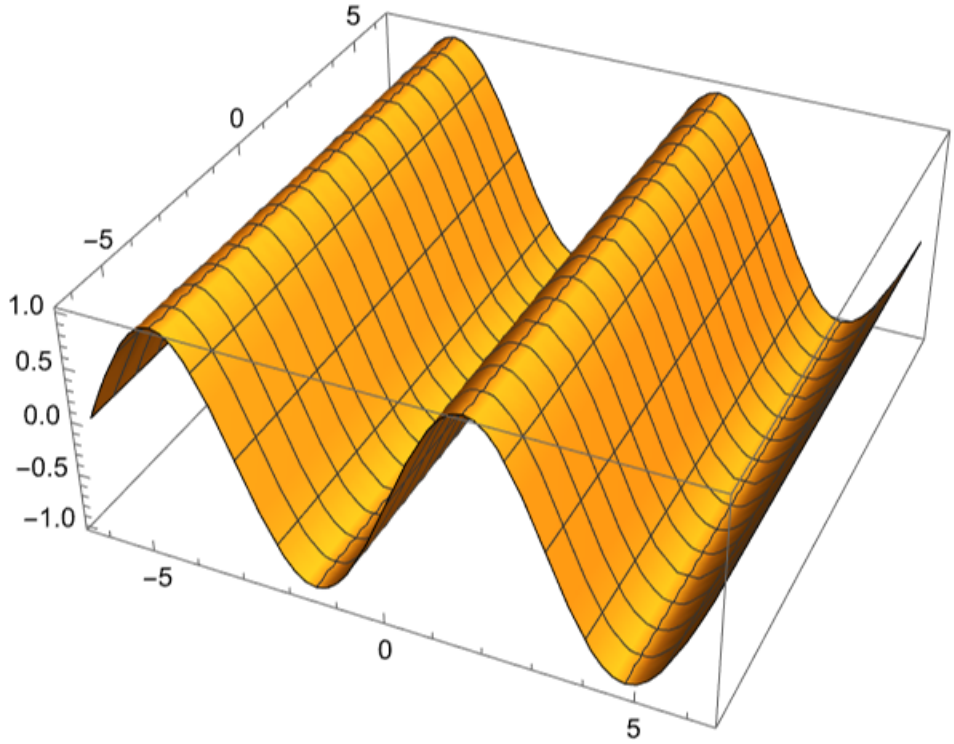


Figure 7: Graph of $z = f(x,y) = \sin(x)$.

The next theorems show that all of the limit properties of single-variable functions carry over to multivariable functions. Together with the previous example, this allows us to calculate limits of many multivariable functions.

Theorem 1.9. *The limit of a multivariable function, if it exists, is unqiue.*

Theorem 1.10 (Limit laws for multivariable functions)**.** *Suppose $f(x,y)$ and $g(x,y)$ are defined on the same open set containing (x_0,y_0), and that*

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y)\to(x_0,y_0)} g(x,y) = M.$$

Then

(i) $\lim_{(x,y)\to(x_0,y_0)} cf(x,y) = cL$ *for any* $c \in \mathbb{R}$*;*

(ii) $\lim_{(x,y)\to(x_0,y_0)} (f(x,y)+g(x,y)) = L+M$*;*

(iii) $\lim_{(x,y)\to(x_0,y_0)} (f(x,y)g(x,y)) = LM$*;*

(iv) $\lim_{(x,y)\to(x_0,y_0)} = \frac{f(x,y)}{g(x,y)} = \frac{L}{M}$ *whenever* $M \neq 0$*.*

Example 1.11. To evaluate $\lim_{(x,y)\to(1,2)}(x^2y^3 - x^3y^2 + 3x + 2y)$ using Theorems 1.10 and 1.5, we have

$$\begin{aligned}
&\lim_{(x,y)\to(1,2)}(x^2y^3 - x^3y^2 + 3x + 2y) = \lim_{(x,y)\to(1,2)}(x^2y^3) + \lim_{(x,y)\to(1,2)}(-x^3y^2) + \lim_{(x,y)\to(1,2)}(3x+2y)\\
&= \lim_{(x,y)\to(1,2)}x^2 \cdot \lim_{(x,y)\to(1,2)}y^3 - \lim_{(x,y)\to(1,2)}x^3 \cdot \lim_{(x,y)\to(1,2)}y^2 + 3\lim_{(x,y)\to(1,2)}x + 2\lim_{(x,y)\to(1,2)}y\\
&= (\lim_{(x,y)\to(1,2)}x)^2 \cdot (\lim_{(x,y)\to(1,2)}y)^3 - (\lim_{(x,y)\to(1,2)}x)^3 \cdot (\lim_{(x,y)\to(1,2)}y)^2 + 3\lim_{(x,y)\to(1,2)}x + 2\lim_{(x,y)\to(1,2)}y\\
&= (1)^2(2)^3 - (1)^3(2)^2 + 3(1) + 2(2)\\
&= 8 - 4 + 3 + 4\\
&= 11.
\end{aligned}$$

Definition 1.12 (Continuous function)**.** A multivariable function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ is *continuous at* **a** if

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = f(\mathbf{a}).$$

The function f is *continuous on* U if f is continuous at every point in U.

It follows immediately from Theorem 1.10 that that sums, products, and quotients of continuous functions are continuous on their domains.

It follows from Theorem 1.5 that $f(x,y) = x, g(x,y) = y$, and $h(x,y) = c$ are continuous. Since any polynomial, e.g.,

$$f(x,y) = x^4 + 5x^3y^2 + 6xy^4 - 7y + 6$$

can be built out of these by multiplication and addition, all polynomials are continuous on $\mathbb{R}^2$. Likewise, any rational function is continuous on its domain because it is a quotient of continuous functions. E.g.,

$$g(x,y) = \frac{2xy+1}{x^2+y^2}.$$

Example 1.13. The function $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$ is continuous everywhere except at $(0,0)$.

Exercise 1.14. Where is the function

$$g(x,y) = \begin{cases} \frac{x^4-x^2y^2}{x+y}, & \text{if } (x,y) \neq (0,0),\\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

continuous?

Proof. Since a rational function is continuous everywhere it is defined, $g(x,y)$ is continuous for all $(x,y) \neq (0,0)$. Since for $(x,y) \neq (0,0)$

$$\frac{x^4 - x^2y^2}{x+y} = \frac{x^2(x^2-y^2)}{x+y} = \frac{x^2(x+y)(x-y)}{x+y} = x^2(x-y),$$

we have

$$\lim_{(x,y)\to(0,0)} \frac{x^4 - x^2y^2}{x+y} = \lim_{(x,y)\to(0,0)} x^2(x-y) = 0 = g(0,0).$$

Thus, $g(x,y)$ is continuous everywhere. □

Here is another useful theorem.

Theorem 1.15. *A composition of continuous functions is continuous.*

Example 1.16. Since $f(x,y) = \sin(x^2+y^2)$ is a composition of continuous functions,

$$\lim_{(x,y)\to(\frac{\sqrt{\pi}}{2},\frac{\sqrt{\pi}}{2})} \sin(x^2+y^2) = \sin(\frac{\pi}{4}+\frac{\pi}{4}) = \sin(\frac{\pi}{2}) = 1.$$

The proofs of all of these statements all follow from the Triangle Inequality, and are exactly the same as in the one-variable case once we establish that the norm $||\mathbf{x}-\mathbf{y}||$ satisfies the Triangle Inequality, which we now turn to.

Proof of Theorem 1.10

The proof of Theorem 1.10 is essentially the same as the proof of the corresponding theorem for single-variable functions. The latter followed from the triangle inequality, which states that for all $x, y \in \mathbb{R}$,

$$|x+y| \leq |x| + |y|.$$

For example, if $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, then to prove that $\lim_{x\to a}(f(x)+g(x)) = L+M$, let $\varepsilon > 0$ and note that there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that $0 < |x-a| < \delta_1$ implies $|f(x)-L| < \frac{\varepsilon}{2}$ and $0 < |x-a| < \delta_2$ implies $|g(x)-M| < \frac{\varepsilon}{2}$. It follows that, for $0 < |x-a| < \delta := \min\{\delta_1, \delta_2\}$, we have

$$|f(x)+g(x)-(L+M)| = |f(x)-L+g(x)-M| \leq |f(x)-L| + |g(x)-M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So, if we can prove that $||\mathbf{x}+\mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we are done. To do so, we need the following lemma.

Lemma 1.17 (Cauchy-Schwarz inequality)**.** *For all* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|\mathbf{x}\cdot\mathbf{y}| \leq ||\mathbf{x}||\,||\mathbf{y}||.$$

Proof. For every $\lambda \in \mathbb{R}$, we have $||\mathbf{x}-\lambda\mathbf{y}||^2 \geq 0$. If $||\mathbf{x}-\lambda\mathbf{y}||^2 = 0$, then $\mathbf{x} = \lambda\mathbf{y}$ and it follows that

$$\begin{aligned}(\mathbf{x}\cdot\mathbf{y})^2 &= (\mathbf{x}\cdot\lambda\mathbf{x})^2 \\ &= (\lambda||\mathbf{x}||^2)^2 \\ &= \lambda^2||\mathbf{x}||^4 \\ &= ||\mathbf{x}||^2||\mathbf{y}||^2\end{aligned}$$

and therefore

$$|\mathbf{x}\cdot\mathbf{y}| = ||\mathbf{x}||\,||\mathbf{y}||.$$

Otherwise

$$\begin{aligned} 0 &< ||\mathbf{x}-\lambda\mathbf{y}||^2 \\ &= (\mathbf{x}-\lambda\mathbf{y})\cdot(\mathbf{x}-\lambda\mathbf{y}) \\ &= \lambda^2||\mathbf{y}||^2 - 2\lambda\,\mathbf{x}\cdot\mathbf{y} + ||\mathbf{x}||^2 \end{aligned}$$

Since the quadratic expression $\lambda^2||\mathbf{y}||^2 - 2\lambda\,\mathbf{x}\cdot\mathbf{y} + ||\mathbf{x}||^2$ has no real zeros, it must have a negative discriminant:

$$4(\mathbf{x}\cdot\mathbf{y})^2 - 4||\mathbf{x}||^2||\mathbf{y}||^2 < 0$$

which implies

$$(\mathbf{x}\cdot\mathbf{y})^2 < ||\mathbf{x}||^2||\mathbf{y}||^2$$

and therefore

$$\sqrt{(\mathbf{x}\cdot\mathbf{y})^2} = |\mathbf{x}\cdot\mathbf{y}| < ||\mathbf{x}||\,||\mathbf{y}||.$$

This proves that, in all cases,

$$|\mathbf{x}\cdot\mathbf{y}| < ||\mathbf{x}||\,||\mathbf{y}||,$$

completing the proof. □

Proposition 1.18 (Triangle Inequality)**.** *For all* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$||\mathbf{x}+\mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||.$$

Proof. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\begin{aligned} ||\mathbf{x}+\mathbf{y}||^2 &= (\mathbf{x}+\mathbf{y})^2 \\ &= ||\mathbf{x}||^2 + ||\mathbf{y}||^2 + 2\mathbf{x}\cdot\mathbf{y} \\ &\leq ||\mathbf{x}||^2 + ||\mathbf{y}||^2 + 2|\mathbf{x}\cdot\mathbf{y}| \\ &\leq ||\mathbf{x}||^2 + ||\mathbf{y}||^2 + 2||\mathbf{x}||\,||\mathbf{y}|| \qquad \text{(by the Cauchy-Schwarz Inequality)} \\ &= (||\mathbf{x}|| + ||\mathbf{y}||)^2, \end{aligned}$$

and therefore

$$||\mathbf{x}+\mathbf{y}| \leq ||\mathbf{x}|| + ||\mathbf{y}||.$$

□

Technique 2: Factor and Cancel, Conjugate and Cancel

Exercise 1.19. Compute $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^4-y^4}{x^2+y^2}$.

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{x^4-y^4}{x^2+y^2} &= \lim_{(x,y)\to(0,0)} \frac{(x^2+y^2)(x^2-y^2)}{x^2+y^2} \\ &= \lim_{(x,y)\to(0,0)} (x^2-y^2) \\ &= 0. \end{aligned}$$

Example 1.20. Consider the function $f(x,y) = \frac{x^2-xy}{\sqrt{x}-\sqrt{y}}$, which is defined on the complement of the line $y = x$ in $\mathbb{R}^2$. We see that

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}} &= \lim_{(x,y)\to(0,0)} \frac{x(x-y)}{\sqrt{x}-\sqrt{y}} \\ &= \lim_{(x,y)\to(0,0)} \frac{x(\sqrt{x}+\sqrt{y})(\sqrt{x}-\sqrt{y})}{\sqrt{x}-\sqrt{y}} \\ &= \lim_{(x,y)\to(0,0)} x(\sqrt{x}+\sqrt{y}) \\ &= 0. \end{aligned}$$

Technique 3: Squeeze Theorem

Theorem 1.21 (Squeeze Theorem)**.** *If*

(i) $g(x,y) \leq f(x,y) \leq h(x,y)$ *for all* (x,y) *in an open interval containing* (a,b)*, except perhaps at* (a,b) *itself, and*

(ii) $\lim_{(x,y)\to(a,b)} g(x,y) = \lim_{(x,y)\to(a,b)} h(x,y) = L \in \mathbb{R}$,

then $\lim_{(x,y)\to(a,b)} f(x,y) = L$.

The proof is essentially the same as for functions of one-variable.

Example 1.22. Consider the limit $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2\sin^2 y}{x^2+y^2}$. Note that $\dfrac{x^2\sin^2 y}{x^2+y^2} \geq 0$. Note also that $x^2 + y^2 \geq x^2$ and therefore $\dfrac{x^2}{x^2+y^2} \leq 1$ for $(x,y) \neq (0,0)$. We see that

$$0 \leq \frac{x^2\sin^2 y}{x^2+y^2} \leq \sin^2 y$$

for all $(x,y) \neq (0,0)$. Since $\lim_{(x,y)\to(0,0)} \sin^2 y = 0$ and $\lim_{(x,y)\to(0,0)} 0 = 0$, it follows from the Squeeze Theorem that

$$\lim_{(x,y)\to(0,0)} \frac{x^2\sin^2 y}{x^2+y^2} = 0.$$